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December 10, 1965
SUPERSEDING
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FEDERAL SPECIFICATION

FEATHERS, LAND FOWL, (CRUSHED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers one type of crushed chicken feathers.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

V-T-276—Thread, Cotton.
QQ-S-781—Steel, Strapping, Flat.
CCC-C-429—Cloth, Osnaburg, Cotton.
CCC-C-432—Cloth, Cotton, Sheeting, (Unbleached, Bleached, and Dyed).
PPP-F-320—Fiberboard: Sheet, Stock and Cut Shapes.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 148—Classification, Identification, and Testing of Feather Filling Material.
Fed. Std. No. 751—Stitches, Seams, and Stitchings.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Laboratory report approval. Unless otherwise specified (see 6.2), prior to or at the time of submission of bid, the bidder shall submit to the contracting officer a certified copy of a recent laboratory report which contains data (see 4.2) demonstrating that the type of product proposed for delivery has been tested and found to comply with the requirements of this specification. Any of the following types of reports will be satisfactory from the standpoint of this requirement:

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(*a*) An independent or commercial laboratory report.
(*b*) The prospective supplier's own laboratory report.
(*c*) A Governmental laboratory report from a contract within not more than six months of submission of bid.

The purpose of the above requirement is to assist the Government to determine the capability of the bidder to supply feathers meeting all the requirements of this specification. The submission of an acceptable report under this requirement shall not be construed as relieving the supplier from subsequently meeting all requirements of this specification.

3.2 Preproduction sample approval. Unless otherwise specified (see 6.2), before production is commenced, a one-pound sample of the finished product shall be submitted or made ready for the contracting officer or his authorized representative for inspection as specified in 4.3. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all the requirements of this specification.

3.3 Materials. The feathers shall be from domesticated chickens and shall be new (not previously used or reclaimed). The feathers shall be of such composition and quality to yield finished processed feathers meeting the requirements of this specification.

3.3.1 *Preservation of raw feathers.* Not more than 24 hours shall be permitted to elapse between time of plucking and beginning of processing as described in 3.4. Feathers which are not to be processed within 12 hours after plucking shall be preserved by steeping in an acid bath containing approximately 10 ounces of a commercial laundry sour of the fluoride type (sodium acid fluoride, sodium silicofluoride, or combination thereof) in each 100 gallons of water. Feathers which have been held by acid steeping shall be rinsed with clean water before proceeding with the processing described in 3.4.

3.4 Processing.

3.4.1 *Cleaning.* The feathers shall be washed blood-free with the aid of a blood-solubilizing agent, scoured with detergent, and rinsed. The final rinse shall contain a commercial laundry sour of the fluoride type (see 3.3.1). The feathers shall then be dried at sufficiently high temperature to effect sterilizing conditions (see 6.3).

3.4.2 *Crushing.* The feathers shall be processed by a crushing, chopping, or curling machine to alter the original flat configuration of the feathers without removing the quill.

3.5 Finished requirements.

3.5.1 *Acidity (pH).* The finished product shall show a pH value within the limits of 3.5 to 6.0 when tested as specified in 4.6.

3.5.2 *Oxygen number.* The finished product shall show an oxygen number not greater than 12 when tested as specified in 4.6.

3.5.3 *Solvent soluble material.* The solvent soluble material in the finished product shall not exceed 2.5 percent when tested as specified in 4.6.

3.5.4 *Composition.* The proportional composition by weight of the finished product shall conform to the following requirements when tested as specified in 4.6 in accordance with applicable definitions in Fed. Std. No. 148:

Crushed feathers	Minimum	75 percent
Whole feathers (including quill feathers 4 inches or less)	Maximum	25 percent
Quill feathers over 4 inches	Maximum	3 percent
Residual matter	Maximum	5 percent

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless

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otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Laboratory report approval. The data contained in the report submitted in accordance with 3.1 shall be obtained in accordance with test methods specified in 4.6.

4.3 Preproduction sample inspection. The preproduction sample submitted in accordance with 3.2 shall be examined and tested to determine compliance with the requirements of section 3.

4.4 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5 In-process inspection. Inspection shall be conducted to see that the processing is in accordance with the specified requirements. The supplier shall certify conformance to the following requirements:

(*a*) Use of new chicken feathers put into processing within 24 hours from time of plucking (see 3.3 and 3.3.1).

(*b*) Use of specified acid steep if feathers are not put into processing within 12 hours of plucking (see 3.3.1).

(*c*) Conduct of processing as specified (see 3.4).

4.6 Testing of the end item. Testing of the end item shall be performed in accordance with the methods specified in table I. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result. The test methods indicated by number shall be performed as specified in Fed. Std. No. 148. The sample unit for testing shall be a thoroughly mixed 24-ounce composite of the finished product. Not more than 6 ounces shall be taken from any one 50-pound bag (see 5.1.1). For larger bags, samplings shall be from top, center, bottom and opposite sides of the bag, with no more than 12 ounces from any one bag. The material obtained shall be thoroughly mixed in a screened container with a solid base. Care shall be taken in mixing to prevent damaging or compressing of the material. The lot size shall be expressed in terms of pounds each. The sample size (number of sample units) and the acceptance number shall be in accordance with the following:

Lot size (pounds)	Sample size (units)	Acceptance number
3,200 and less	3	0
3,201 through 10,000 ...	5	0
10,001 through 35,000 ..	8	1
35,001 and over	13	1

TABLE I. *Test methods*

Characteristic	Requirement reference	Test method
Acidity (pH)	3.5.1	8
Oxygen number	3.5.2	4
Solvent soluble material .	3.5.3	5
Composition	3.5.4	2[1]

[1] Classification shall be in accordance with applicable definition in Fed. Std. No. 148.

4.7 Examination of preparation for delivery requirements.

4.7.1 *Inspection of fabricated bags and bale-boards.* Fabricated bags and baleboards shall be inspected to determine conformance with the applicable requirements of 5.1 and 5.2. The acceptable quality level (AQL) shall be 4.0 defects per 100 units

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and the inspection level shall be S-2 of MIL-STD-105.

4.7.2 *Examination of the end item.* The end item (completely packaged and packed feathers) shall be inspected in accordance with the examinations and tests specified in 4.7.2.1, 4.7.2.2, and 4.7.2.3.

4.7.2.1 *Weight.* The lot shall be unacceptable if the total net weight of the feathers in the sample is less than the total net weight indicated on the bags. The number of bags to be selected for the weight examination shall be in accordance with level S-4 of MIL-STD-105.

4.7.2.2 *Packaging examination.* The sample unit shall be one bag of feathers. The inspection level shall be S-1 of MIL-STD-105 and the AQL shall be 6.5 defects per hundred units. Defects shall be scored in accordance with the list below.

Defects
Not packaged in accordance with specified packaging level.
Bag not new, or any patch.
Not closed as specified.
Wire tie broken or not tightly turned to accomplish a secure closure.
Any hole, rip, tear, open seam; any weak places in bag which may develop into a hole.
Bag wet.
Marking—missing, incomplete, illegible, incorrect, not accomplished in specified manner, or not in proper location.

4.7.2.3 *Packing examination.* The sample unit shall be one bale or shipping container. The inspection level shall be level I of MIL-STD-105 and the AQL shall be 6.5 defects per hundred units. Defects shall be scored in accordance with the list below:

Defects
Not packed in accordance with specified packing level.
Bags not compressed to specified size.
Does not contain the specified number of bags.
Kraft facing of baling board not on outside.
Any board missing or strap missing.
Baling board or strap too loose or not placed in proper location to a degree where safe delivery of contents may be impaired.
Marking—missing, incomplete, illegible, incorrect, not as specified or not in proper location.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.* Fifty ± 5 pounds of finished feathers shall be packaged in a new (not used or laundered) plain textile bag fabricated from cotton sheeting conforming to type VIII of CCC-C-432 or osnaburg conforming to class 2 of CCC-C-429. Cloths in other widths of equivalent weight and construction will be acceptable.

5.1.1.1 *Bag, construction.* The bags shall have a finished length and width measured flat of approximately 72 by 36 inches and shall be constructed in accordance with the following requirements: The side and bottom shall be seamed with 5 to 6 stitches per inch with either seam and stitching type 401, SSa-2 or 301, SSa-2, conforming to Fed. Std. No. 751. The stitching shall be at least 5/8 inch from the cut edge of the cloth or 1/4 inch from a selvage edge and each row shall be at least 1/8 inch apart. The ends of all stitching, when not caught in other seams or stitching shall be backstitched or backtacked at least 1 inch. The thread used for seaming shall be cotton, machine type IA1 or IB1, unbleached, ticket No. 12, 4 ply conforming to V-T-276.

5.1.1.1.1 *Type* 301 *stitching.* Thread breaks in stitching shall be backstitched not less than 1 inch at each break. Thread tension shall be maintained so that there will be no loose stitching. Thread ends shall be trimmed.

5.1.1.1.2 *Type* 401 *stitching.* Thread breaks in stitching shall be overstitched not less than 1-1/2 inches at each break. Thread tension shall be maintained so that

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there will be no loose stitching. Thread ends shall be trimmed.

5.1.1.2 *Closure.* The closure of the top of the bag, after filling, shall be effected by means of two double loops of 0.072 inch thick galvanized soft iron or steel wire ties of not less than 6 inches long, having a 1/2-inch diameter eye formed at each end. One tie shall be applied at the base of an ear formed by gathering the cloth evenly and the second tie applied at a distance approximately 1 inch from the first tie with the twisted ends of two ties alternated and facing in opposite direction.

5.1.2 *Level C.* Feathers shall be packaged in accordance with the industry's practice.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 *Level A.* Three bags (150 pounds of feathers with permissible tolerances, as specified in 5.1.1) shall be placed between two baling boards and compressed by machine so as to form a bale approximately one-fifth of the original volume. Bales thus formed shall measure approximately 60 by 24 by 20 inches.

5.2.1.1 *Baling boards.* Baling boards shall be not less than 0.075 inch thick solid fiberboard conforming to type SF, class domestic, grade 200 of PPP-F-320, except that at least one face of the board shall be not less than 0.010-inch kraft paperboard. Two such baling boards, size 58 by 36 inches, each with a score line placed 6 inches from each 58-inch edge in such manner that the kraft facing is on the outside when in a folded position, shall be used in each bale; one board on top of bags being baled, and one on the bottom. The boards shall be bent at the score lines thereby forming a fiberboard cap 58 inches in length, 24 inches in width, with 6-inch flanges extended down over the side of the bale.

5.2.1.2 *Strapping.* Not less than eight straps shall be used on each bale; straps shall be affixed girthwise around the sides, top, and bottom of the bale over the baling boards. The two outside straps shall be placed 4 inches in from each end and the intermediate straps placed equidistant between the end bands. All strapping shall be flat steel not less than 3/8 inch wide by 0.020 inch thick conforming to type I, class A of QQ-S-781.

5.2.2 *Level C.* Feathers, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rates for such supplies. Containers shall be in accordance with the rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The crushed land fowl feathers covered by this specification are intended for use as filling material in pillows and other items.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) When laboratory report approval is not required (see 3.1).
(*c*) When preproduction sample is not required (see 3.2).
(*d*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.3 Processing. With reference to the processing requirements of 3.4.1, the following procedure is suggested to yield a satisfactory product:

(*a*) Within 30 minutes after wet plucking the feathers are given a clear water rinse at temperature not over 105° F.

(*b*) Rinsed feathers are then agitated for 20 minutes in water at 95° ± 5° F. to which has been added 1 ounce of blood-solubilizing agent for each 10 gallons of water.

(*c*) Feathers are then scoured with detergent in warm water, and thoroughly rinsed. In the final rinse a commercial sour of prescribed type is added.

(*d*) Feathers are then centrifuged, and dried in a steam heated dryer with means of agitation. The drying cycle is controlled to attain moist heat of 165° F. or higher at the exhaust stack for a minimum of 5 minutes to effect sterilizing conditions.

6.4 Transportation descriptions. The transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Feathers, not otherwise indexed by name. (Specify pounds per cubic feet.)

Carload minimum weights: 15,000 and 18,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Feathers, not otherwise indexed. (Specify pounds per cubic feet.)

Truckload minimum weights: 10,000, 14,000, 18,000 and any quantity, subject to Rule 115, National Motor Freight Classification.

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Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price cents each.

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